

# Chapter 4:

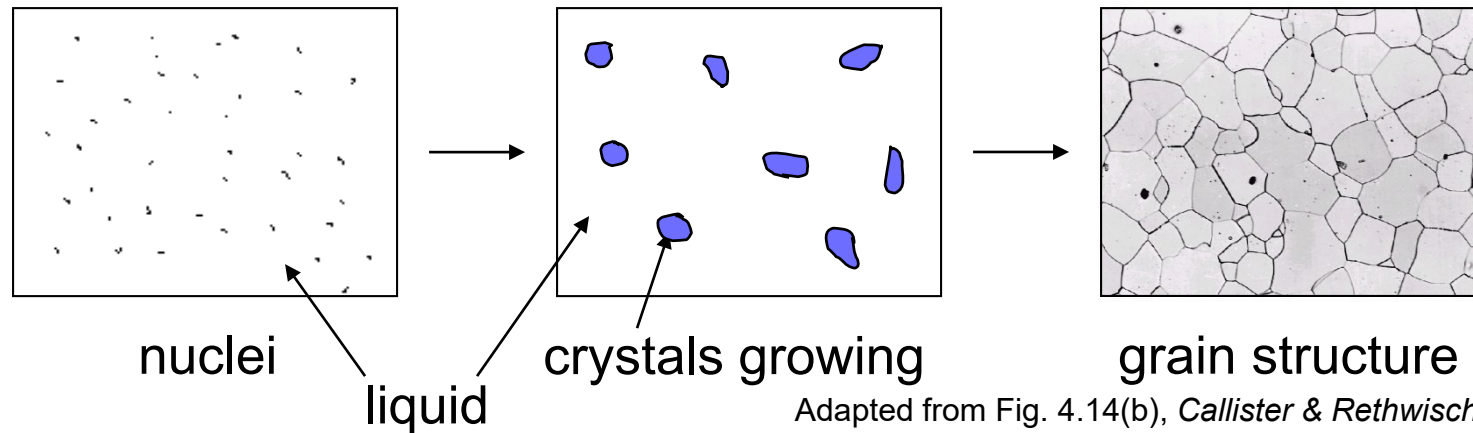
## Imperfections in Solids

### **ISSUES TO ADDRESS...**

- What are the solidification mechanisms?
- What types of defects arise in solids?
- Can the number and type of defects be varied and controlled?
- How do defects affect material properties?
- Are defects undesirable?

# Imperfections in Solids

- **Solidification**- result of casting of molten material
  - 2 steps
    - Nuclei form
    - Nuclei grow to form crystals – grain structure
  - Start with a molten material – all liquid

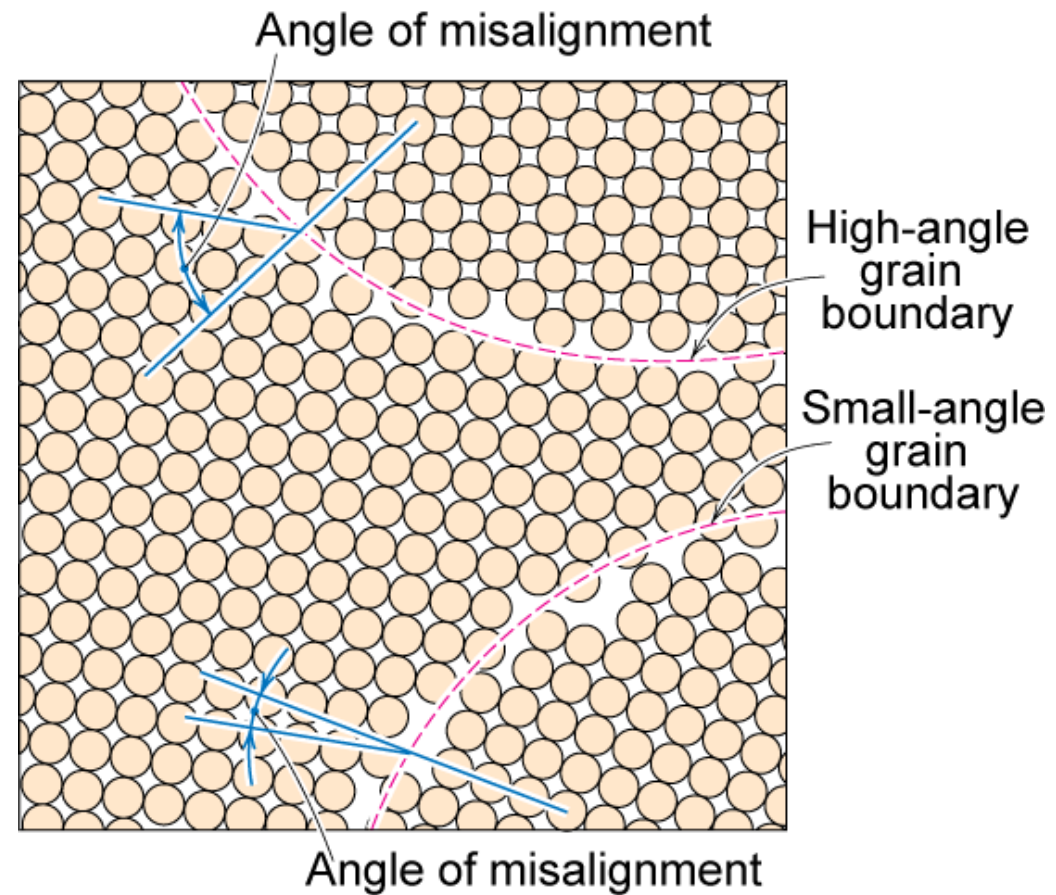


- Crystals grow until they meet each other

# Polycrystalline Materials

## Grain Boundaries

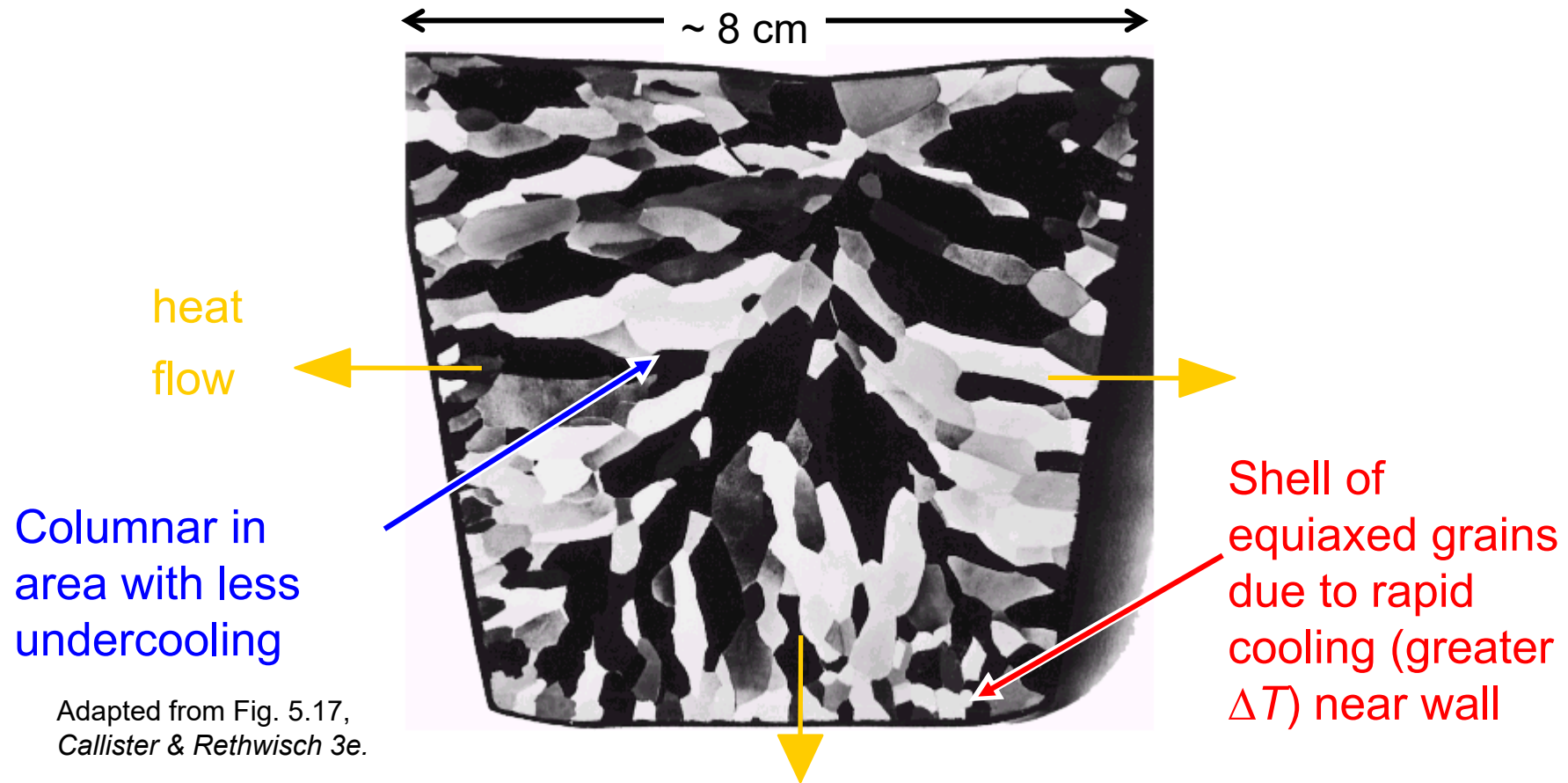
- regions between crystals
- transition from lattice of one region to that of the other
- slightly disordered
- low density in grain boundaries
  - high mobility
  - high diffusivity
  - high chemical reactivity



Adapted from Fig. 4.7,  
*Callister & Rethwisch 8e.*

# Solidification

- Grains can be
- equiaxed (roughly same size in all directions)
  - columnar (elongated grains)



Adapted from Fig. 5.17,  
*Callister & Rethwisch 3e.*

Grain Refiner - added to make smaller, more uniform, equiaxed grains.

# Imperfections in Solids

There is no such thing as a perfect crystal.

- What are these imperfections?
- Why are they important?

Many of the important properties of materials are due to the presence of imperfections.

# Types of Imperfections

- Vacancy atoms
- Interstitial atoms
- Substitutional atoms

Point defects

- Dislocations

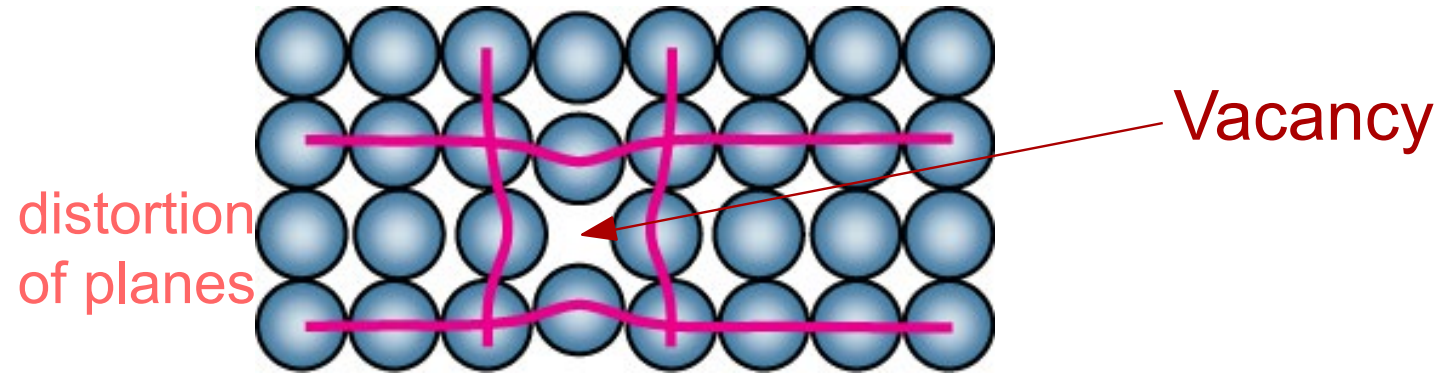
Line defects

- Grain Boundaries

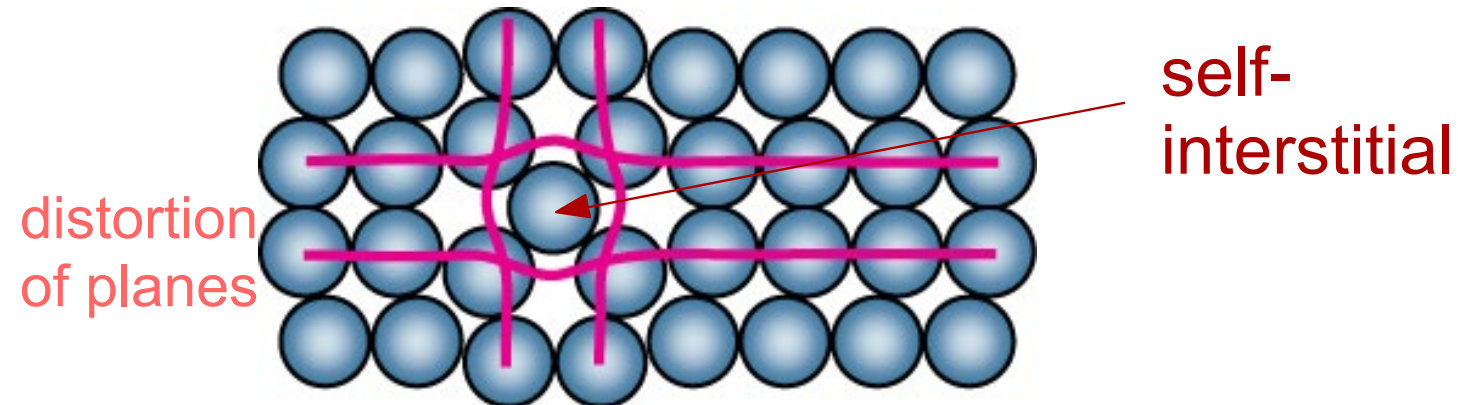
Area defects

# Point Defects in Metals

- **Vacancies:**  
-vacant atomic sites in a structure.



- **Self-Interstitials:**  
-"extra" atoms positioned between atomic sites.

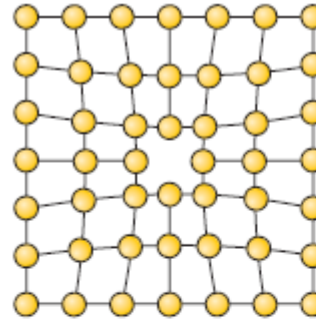


# Point Defects

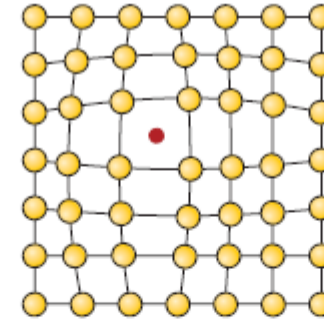
a. Vacancy

b. Interstitial atom

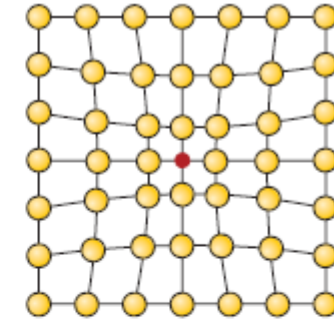
c. Substitutional (small)



(a)



(b)

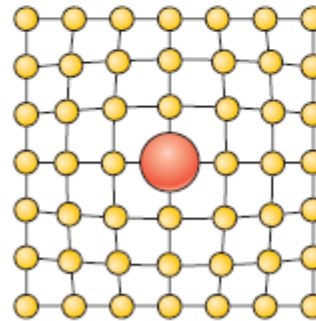


(c)

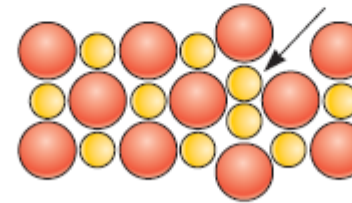
d. Substitutional (large)

e. Frenkel Defect

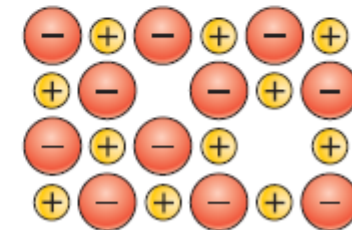
f. Schottky Defect



(d)



(e)



(f)



# Point Defects (interstitial)

An interstitial defect is formed when an **extra atom/ion** is inserted into the crystal structure at a normally **unoccupied position**.

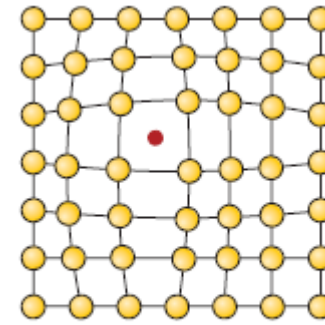
Although smaller than atoms at lattice points, interstitial atoms introduce a **localized stress** in their vicinity.

This stress often **strengthens the material**. For instance, interstitial carbon atoms in an iron lattice strengthen the iron.

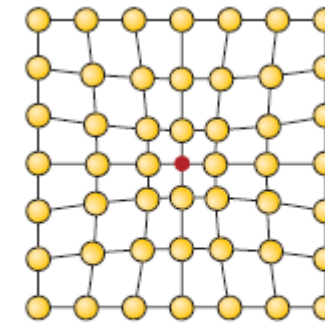
Point defects are **localized disruptions** in otherwise perfect atomic/ionic arrangements in a crystal structure.

Effects of these **disruptions** are often felt over the surrounding region.

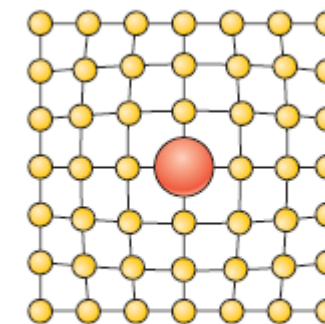
Substitutional atoms often **increase the strength** of metallic materials.



(b)



(c)

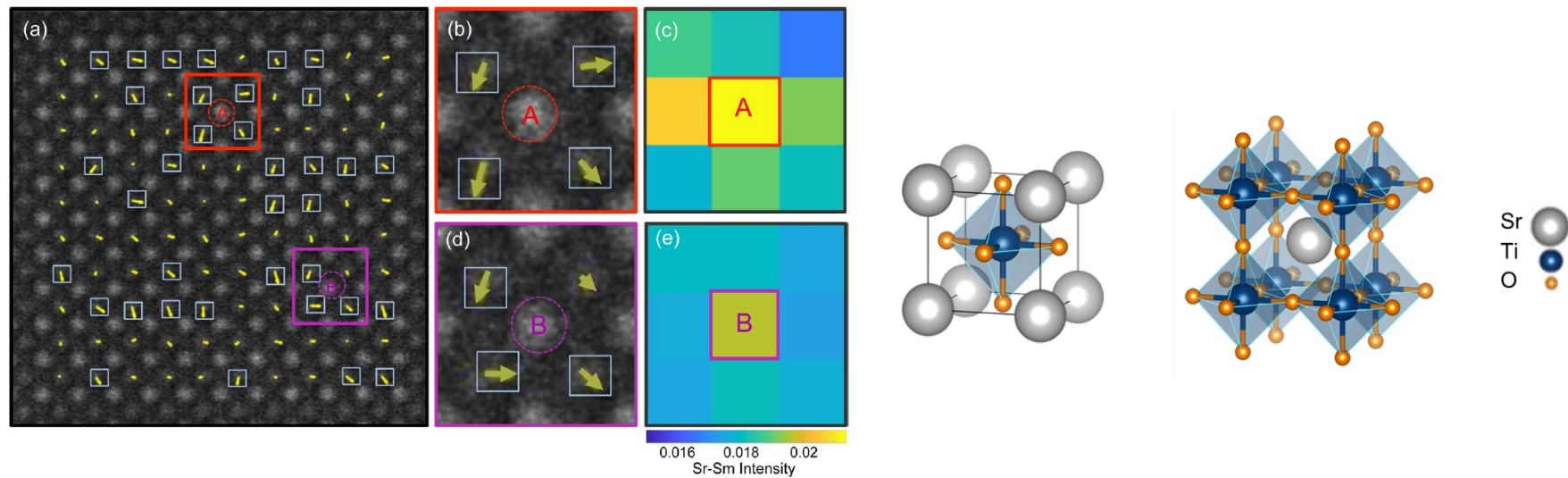


(d)

## Lattice relaxations around individual dopant atoms in SrTiO<sub>3</sub>

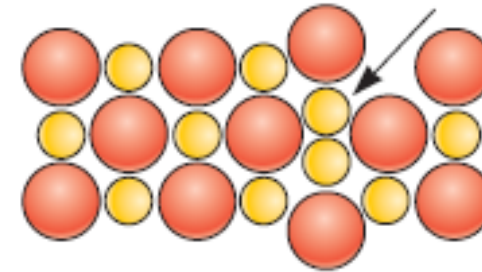
Salva Salmani-Rezaie, Honggyu Kim, Kaveh Ahadi, and Susanne Stemmer<sup>\*</sup>  
*Materials Department, University of California, Santa Barbara, California 93106–5050, USA*

 (Received 3 July 2019; revised manuscript received 22 September 2019; published 8 November 2019)

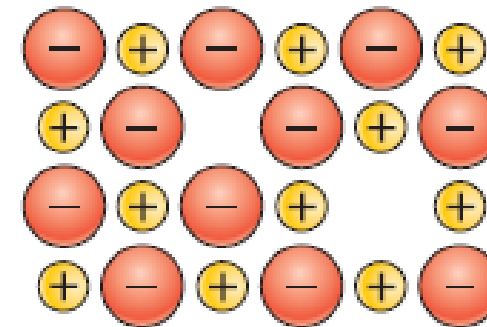


## Other Point Defects

- A Frenkel defect is a **vacancy-interstitial pair** formed when a lattice atom jumps to an interstitial site, leaving behind a vacancy.
- A Schottky defect is unique to ionic materials. To preserve electrical neutrality in ionic materials, for any **vacancies**, a stoichiometric number of **cations & anions** must be removed from the lattice.



(e)



(f)

# Equilibrium Concentration: Point Defects

- Equilibrium concentration varies with temperature!

No. of defects  $\rightarrow N_v$

No. of potential defect sites  $\rightarrow N$

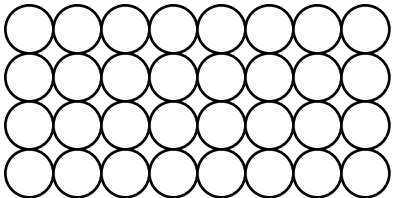
Activation energy  $\rightarrow Q_v$

Boltzmann's constant  $\rightarrow k$

Temperature  $\rightarrow T$

$$\frac{N_v}{N} = \exp\left(\frac{-Q_v}{kT}\right)$$

Each lattice site is a potential vacancy site



(1.38 x 10<sup>-23</sup> J/atom-K)  
(8.62 x 10<sup>-5</sup> eV/atom-K)

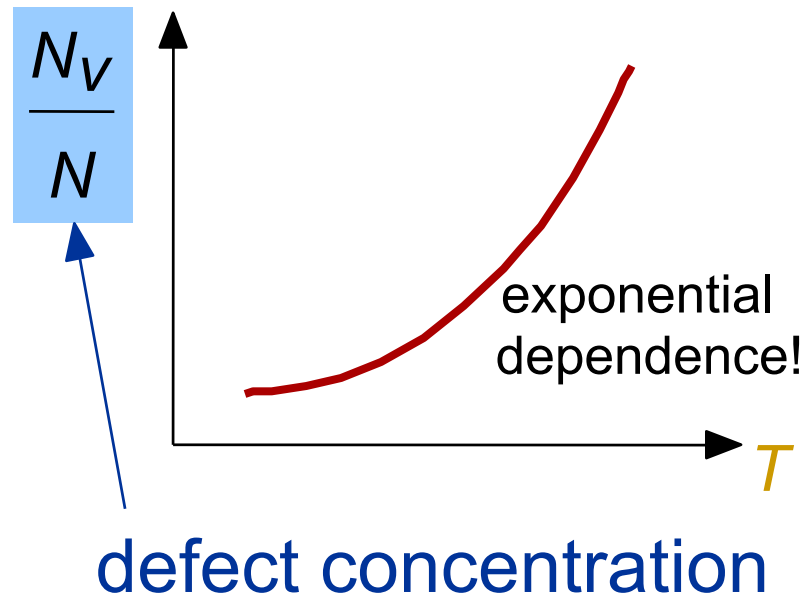
# Measuring Activation Energy

Vacancy density doubles with increasing the temperature from 300 K to 400 K.

What is the activation energy assuming it is not changing with temperature?

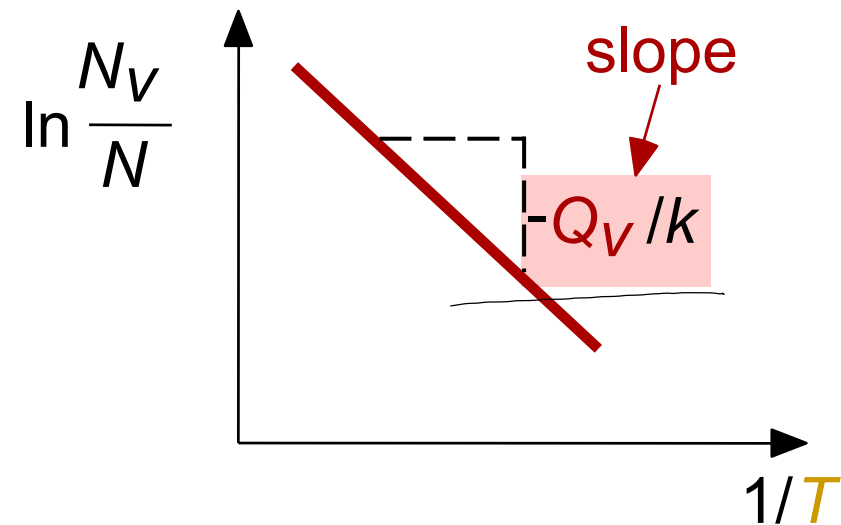
# Measuring Activation Energy

- We can get  $Q_v$  from an experiment.
- Measure this...



$$\frac{N_v}{N} = \exp\left(\frac{-Q_v}{kT}\right)$$

- Replot it...



# Estimating Vacancy Concentration

- Find the equil. # of vacancies in 1 m<sup>3</sup> of Cu at 1000°C.
- Given:

$$\rho = 8.4 \text{ g/cm}^3 \quad A_{\text{Cu}} = 63.5 \text{ g/mol}$$

$$Q_V = 0.9 \text{ eV/atom} \quad N_A = 6.02 \times 10^{23} \text{ atoms/mol}$$

$$\frac{N_V}{N} = \exp\left(\frac{-Q_V}{kT}\right) = 2.7 \times 10^{-4}$$

For 1 m<sup>3</sup>,  $N = \rho \times \frac{N_A}{A_{\text{Cu}}} \times 1 \text{ m}^3 = 8.0 \times 10^{28} \text{ sites}$

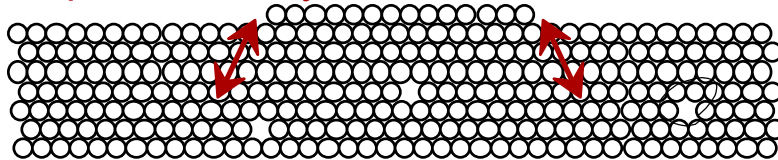
- Answer:

$$N_V = (2.7 \times 10^{-4})(8.0 \times 10^{28}) \text{ sites} = 2.2 \times 10^{25} \text{ vacancies}$$

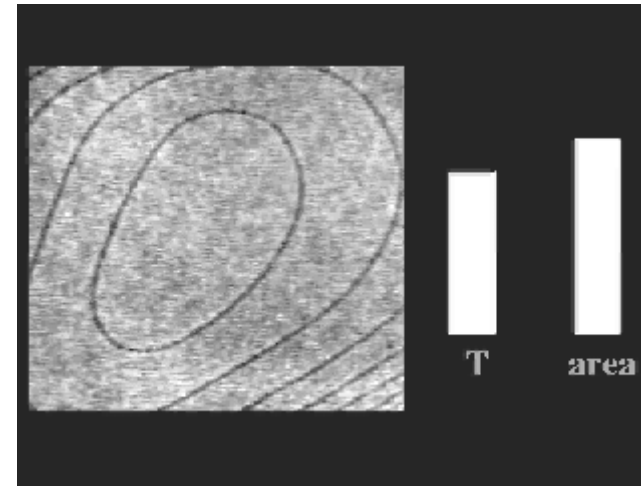
# Observing Equilibrium Vacancy Conc.

- Low energy electron microscope view of a (110) surface of NiAl.
- Increasing temperature causes surface island of atoms to grow.
- Why? The equil. vacancy conc. increases via atom motion from the crystal to the surface, where they join the island.

Island grows/shrinks to maintain equil. vacancy conc. in the bulk.



Click once on image to start animation



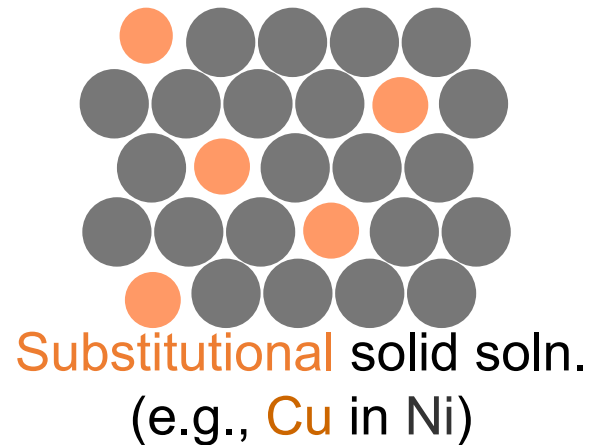
Reprinted with permission from Nature (K.F. McCarty, J.A. Nobel, and N.C. Bartelt, "Vacancies in Solids and the Stability of Surface Morphology", Nature, Vol. 412, pp. 622-625 (2001). Image is 5.75  $\mu\text{m}$  by 5.75  $\mu\text{m}$ .) Copyright (2001) Macmillan Publishers, Ltd.



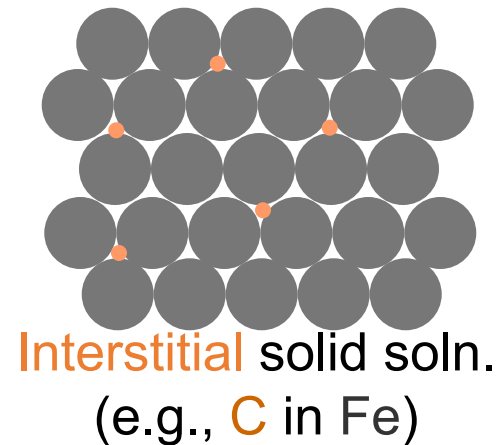
# Imperfections in Metals (i)

Two outcomes if impurity (B) added to host (A):

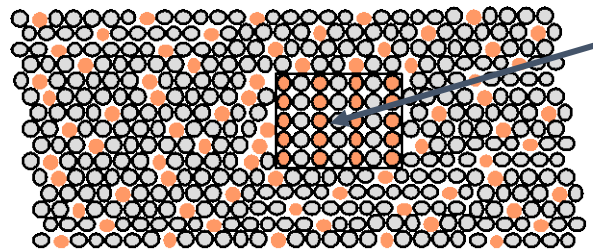
- **Solid solution** of **B** in A (i.e., random dist. of point defects)



OR



- Solid solution of **B** in A plus particles of a new phase (usually for a larger amount of B)



Second phase particle  
-- different **composition**  
-- often different structure.

# Imperfections in Metals (ii)

## Conditions for substitutional solid solution (S.S.)

- W. Hume – Rothery rule

- 1.  $\Delta r$  (atomic radius) < 15%
- 2. Proximity in periodic table
  - i.e., similar electronegativities
- 3. Same crystal structure for pure metals
- 4. Valency
  - All else being equal, a metal will have a greater tendency to dissolve a metal of higher valency than one of lower valency

# Imperfections in Metals (iii)

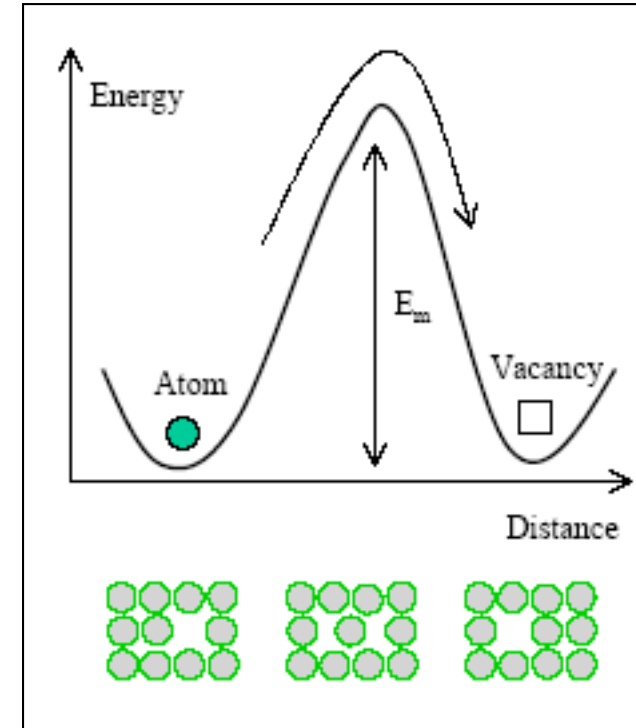
## Application of Hume–Rothery rules – Solid Solutions

| <i>Element</i> | <i>Atomic<br/>Radius<br/>(nm)</i> | <i>Crystal<br/>Structure</i> | <i>Electro-<br/>nega-<br/>tivity</i> | <i>Valence</i> |
|----------------|-----------------------------------|------------------------------|--------------------------------------|----------------|
| Cu             | 0.1278                            | FCC                          | 1.9                                  | +2             |
| C              | 0.071                             |                              |                                      |                |
| H              | 0.046                             |                              |                                      |                |
| O              | 0.060                             |                              |                                      |                |
| Ag             | 0.1445                            | FCC                          | 1.9                                  | +1             |
| Al             | 0.1431                            | FCC                          | 1.5                                  | +3             |
| Co             | 0.1253                            | HCP                          | 1.8                                  | +2             |
| Cr             | 0.1249                            | BCC                          | 1.6                                  | +3             |
| Fe             | 0.1241                            | BCC                          | 1.8                                  | +2             |
| Ni             | 0.1246                            | FCC                          | 1.8                                  | +2             |
| Pd             | 0.1376                            | FCC                          | 2.2                                  | +2             |
| Zn             | 0.1332                            | HCP                          | 1.6                                  | +2             |

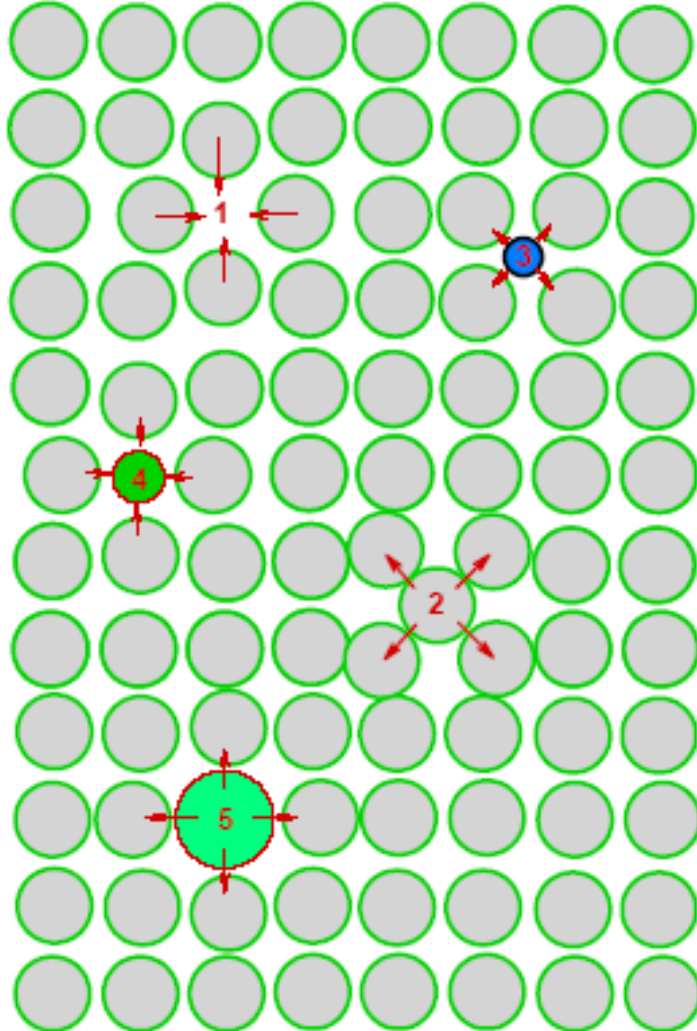
Table on p. 118, *Callister & Rethwisch 8e.*

# Point Defects: Vacancies & Interstitials

- Most common defects in crystalline solids are point defects.
- At high temperatures, atoms frequently and randomly change their positions leaving behind empty lattice sites.
- In general, **diffusion** (mass transport by atomic motion) - can only occur because of vacancies.



# Point Defects: Vacancies & Interstitials



Schematic representation of a variety of point defects:

(1) vacancy;

(2) self-interstitial;  $E_i > E_v$ , so ?

(3) interstitial impurity; less distortion caused

(4,5) substitutional impurities

The arrows represent the local stresses introduced by the point defects.

# Impurities / Solid Solutions

- Impurities are atoms which are different from the host/matrix
- All solids in nature contain some level of impurity
  - Very pure metals 99.9999%
  - For Cu how much does that make ?  $\sim 1$  impurity per 100 atoms
- Impurities may be introduced intentionally or unintentionally.
  - Examples: carbon added in small amounts to iron makes steel, which is stronger than pure iron. Boron is added to silicon change its electrical properties. Pt and Cu are added to Gold to make it stronger, also!
- Alloys - deliberate mixtures of metals
  - Example: sterling silver is 92.5% silver – 7.5% copper alloy. Stronger than pure silver.

# Solid Solutions

**Solid solutions are made of a host (the solvent or matrix) which dissolves the minor component (solute). The ability to dissolve is called solubility.**

- **Solvent:** in an alloy, the element or compound present in greater amount
- **Solute:** in an alloy, the element or compound present in lesser amount

## **Solid Solution:**

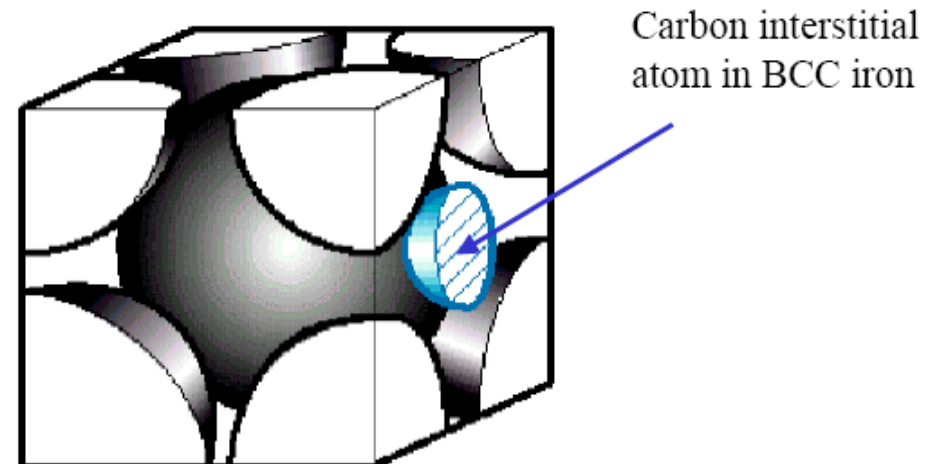
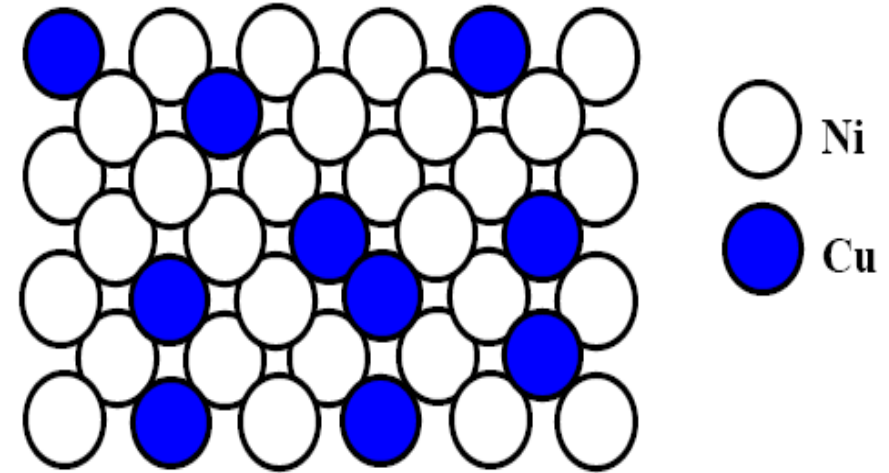
- homogeneous
- maintain crystal structure
- contain randomly dispersed impurities (substitutional or interstitial)

**Second Phase:** while solute atoms are being added, new compounds / structures may form beyond solubility limit, or solute forms local *precipitates* (discussed in Chapter 9)

- Nature of the impurities their concentration, reactivity, temperature and pressure, etc decides the formation of solid solution or a second phase.

# Factors affecting Solid Solubility

- *Atomic size factor* - atoms need to “fit” => solute and solvent atomic radii should be within ~ 15%
- *Crystal structures* - solute and solvent should be crystallize in the same structure
- *Electronegativities* - solute and solvent should have comparable electronegativities, otherwise new inter-metallic phases are encouraged
- *Valency* - generally more solute goes into solution when it has higher valency than solvent





# Impurities in Solids

- Specification of composition

- weight percent

$$C_1 = \frac{m_1}{m_1 + m_2} \times 100$$

$m_1$  = mass of component 1

- atom percent

$$C'_1 = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100$$

$n_{m1}$  = number of moles of component 1

# COMPOSITION

Definition: Amount of impurity (B) and host (A)  
in the **system**.

Two descriptions:

- Weight % (solid solutions)
- Atom % (atomic level)

$$C_B = \frac{\text{mass of B}}{\text{total mass}} \times 100$$

$$C'_B = \frac{\# \text{ atoms of B}}{\text{total \# atoms}} \times 100$$

- Conversion between wt % and at% in an A-B alloy:

$$C_B = \frac{C'_B A_B}{C'_A A_A + C'_B A_B} \times 100$$

$$C'_B = \frac{C_B / A_B}{C_A / A_A + C_B / A_B}$$

- Basis for conversion:

$$\text{mass of B} = \text{moles of B} \times A_B$$

$$\text{mass of A} = \text{moles of A} \times A_A$$

atomic weight of B

atomic weight of A

# Line Defects

## Dislocations:

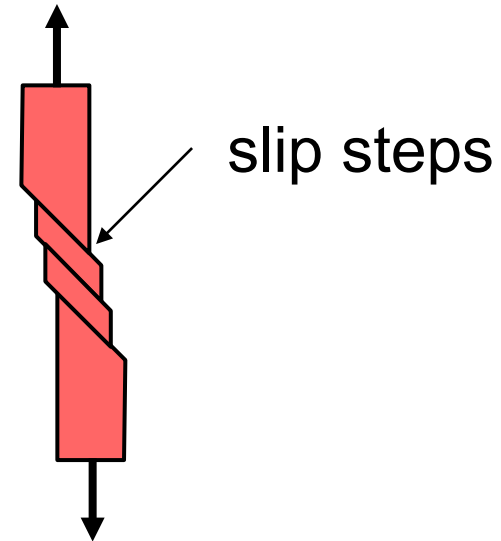
- are line defects,
- slip between crystal planes result when dislocations move,
- produce permanent (plastic) deformation.

## Schematic of Zinc (HCP):

- before deformation

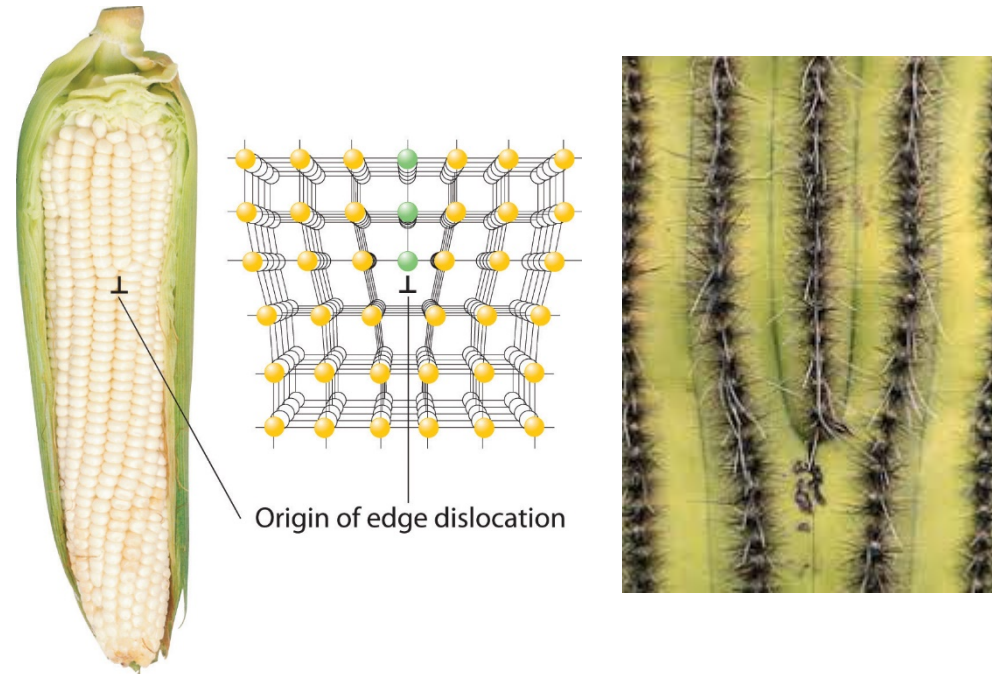


- after tensile elongation



# Dislocations

- **Dislocations are linear defects:** the interatomic bonds are significantly distorted only in the immediate vicinity of the dislocation line. This area is called the **dislocation core**.
- Dislocations also create small elastic deformations of the lattice at large distances.



# Imperfections in Solids

## Edge Dislocation

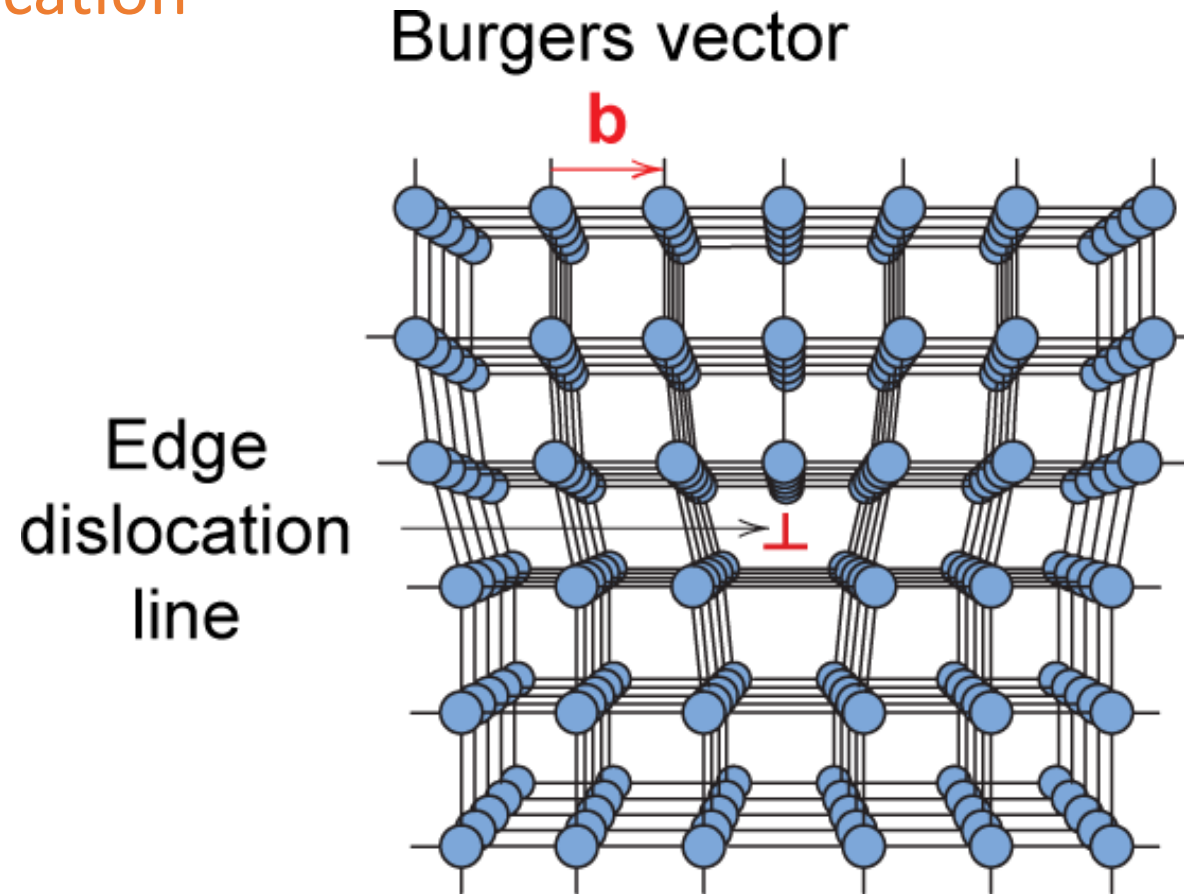
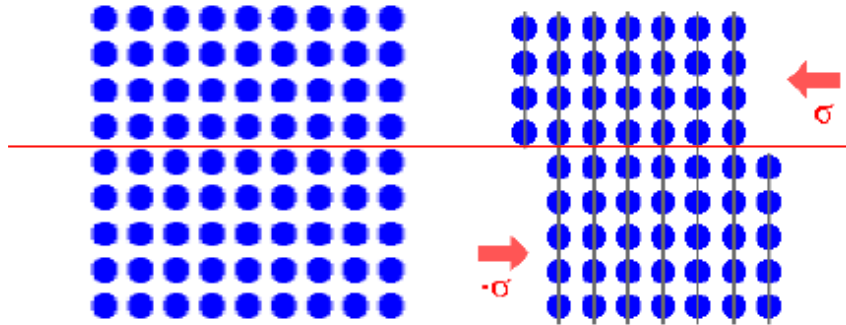


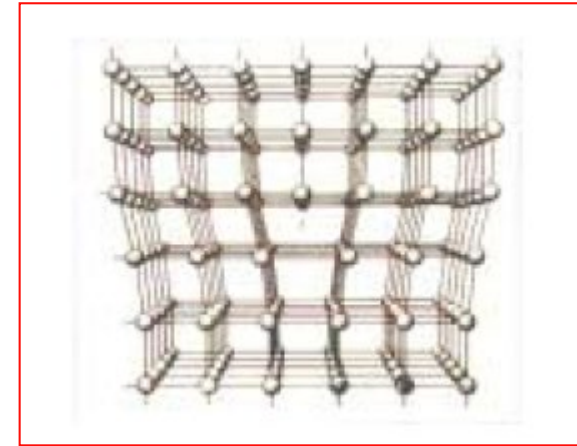
Fig. 4.3, Callister & Rethwisch 8e.

# Issue at Hand



## BASIC THINKING:

The energy required to deform a crystal in the manner described above will require the breaking of all of the bonds on one plane of atoms! However, measured forces required to induce similar *deformations* in single crystals were much lower. How come?



## DISLOCATIONS !

The explanation came in 1934 when Orowan, Polanyi, and Taylor postulated the existence of dislocations\* in crystal structures.

Dislocations can be visualized as an extra lattice plane inserted in the crystal, but not extending through all of the crystal but ending in the dislocation line (HALF - PLANE).

Dislocation motion allows for the slip – plastic deformation- of crystals.

\*Volterra and Timpe in 1900's !

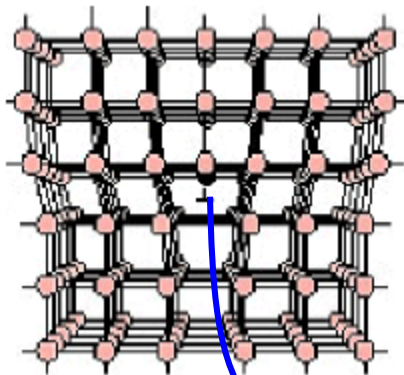
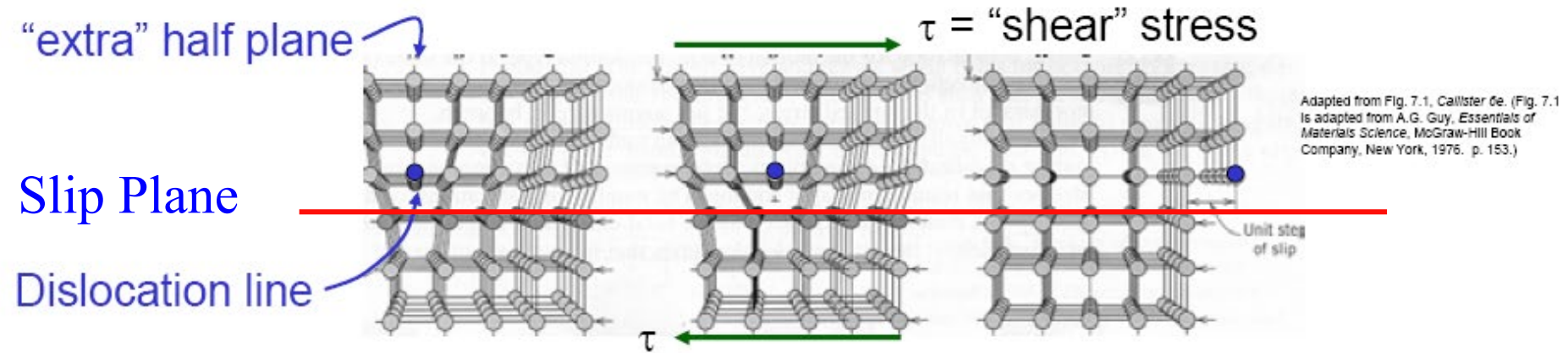
# Line Defects

## Linear Defects (**Dislocations**)

- Are one-dimensional defects around which atoms are misaligned
- **Edge dislocation:**
  - extra half-plane of atoms inserted in a crystal structure
  - **b** perpendicular ( $\perp$ ) to dislocation line
- **Screw dislocation:**
  - spiral planar ramp resulting from shear deformation
  - **b** parallel ( $\parallel$ ) to dislocation line

**Burger's vector, **b**:** measure of lattice distortion

# Dislocation Motion

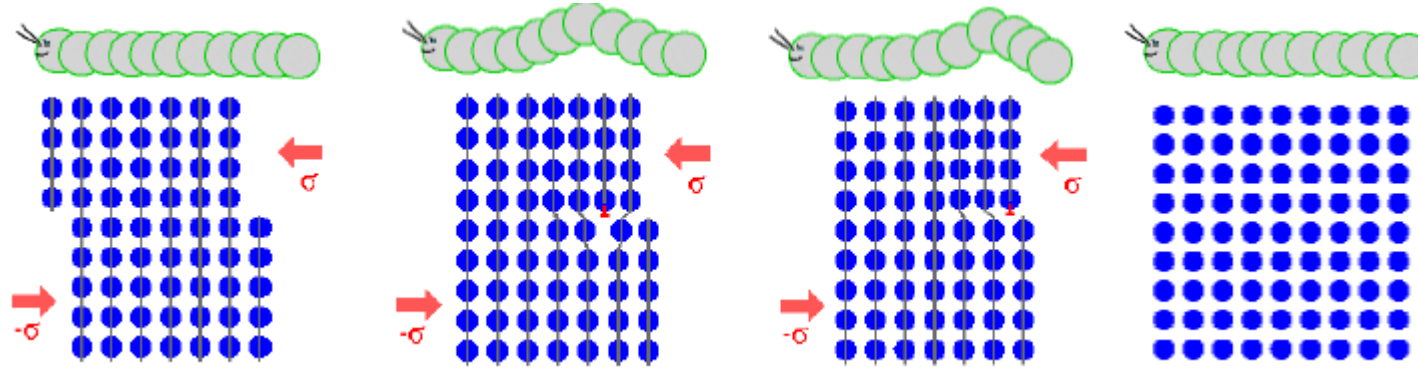


DISLOCATION CORE:  
Induces a stress field due to local distortions in the structure

HOW DOES A DISLOCATION MOVE ?



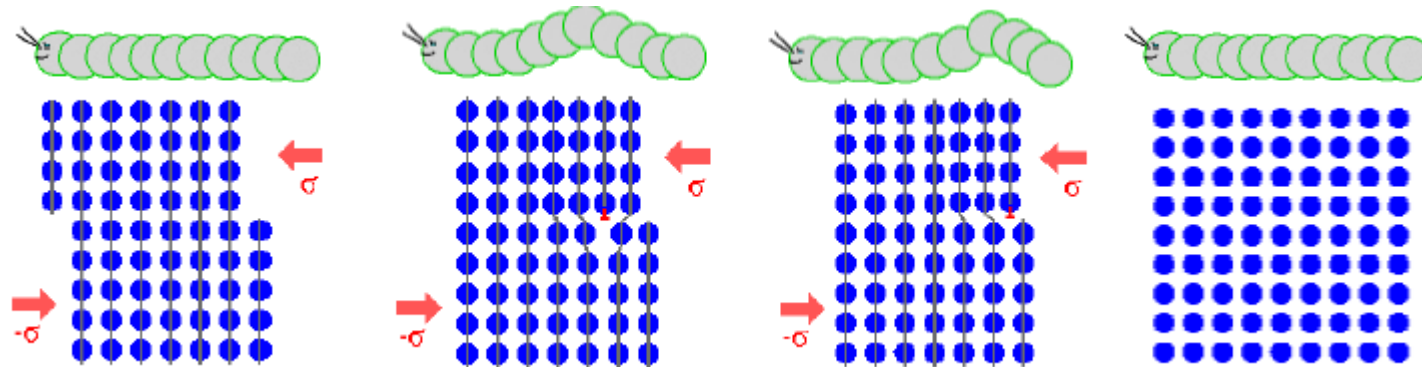
# DISLOCATIONS



- Material *permanently* deforms as dislocation moves through the crystal.
- Bonds break and reform, but *only along the dislocation line* at any point in time, not along the whole plane at once.
- *Dislocation line* separates slipped and unslipped material.

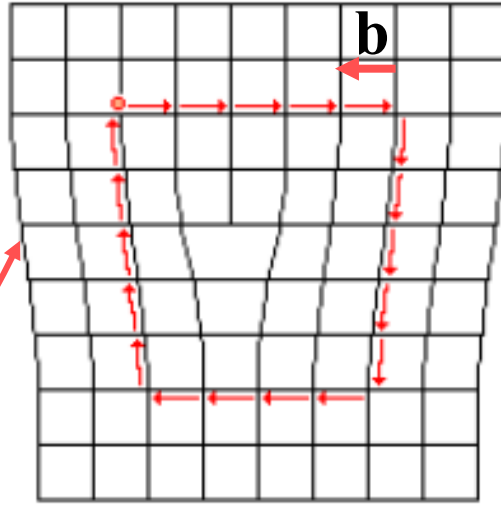
# Dislocation Motion

- Dislocation motion is analogous to movement of a caterpillar:
  - The caterpillar instead of spending a lot of energy to move its entire body at once, moves its forward a bit and creates a *hump*, i.e. a dislocation ! The *hump* then propagates along the caterpillar and moves the caterpillar by a small amount.

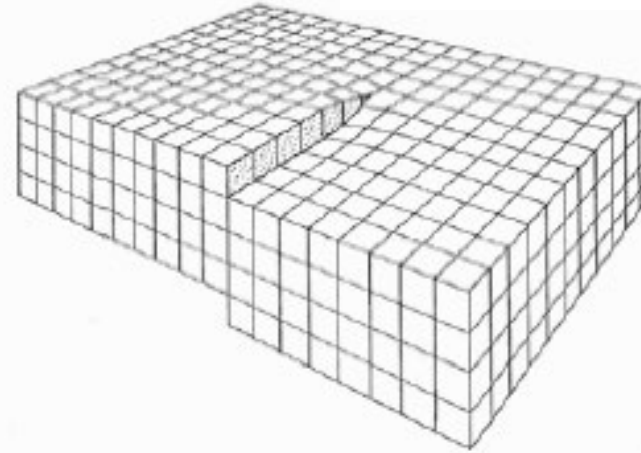


# More about Dislocations

Edge Dislocation



Screw Dislocation

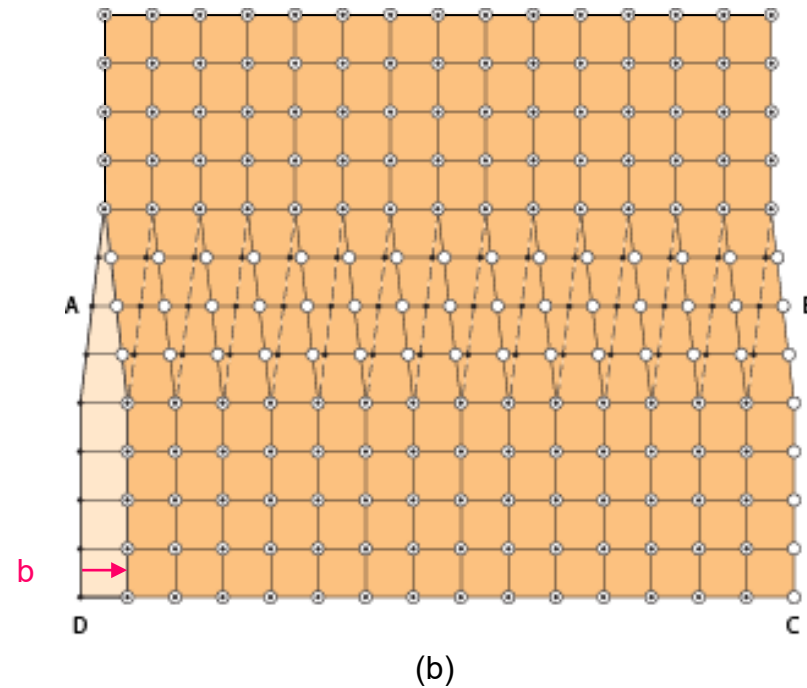
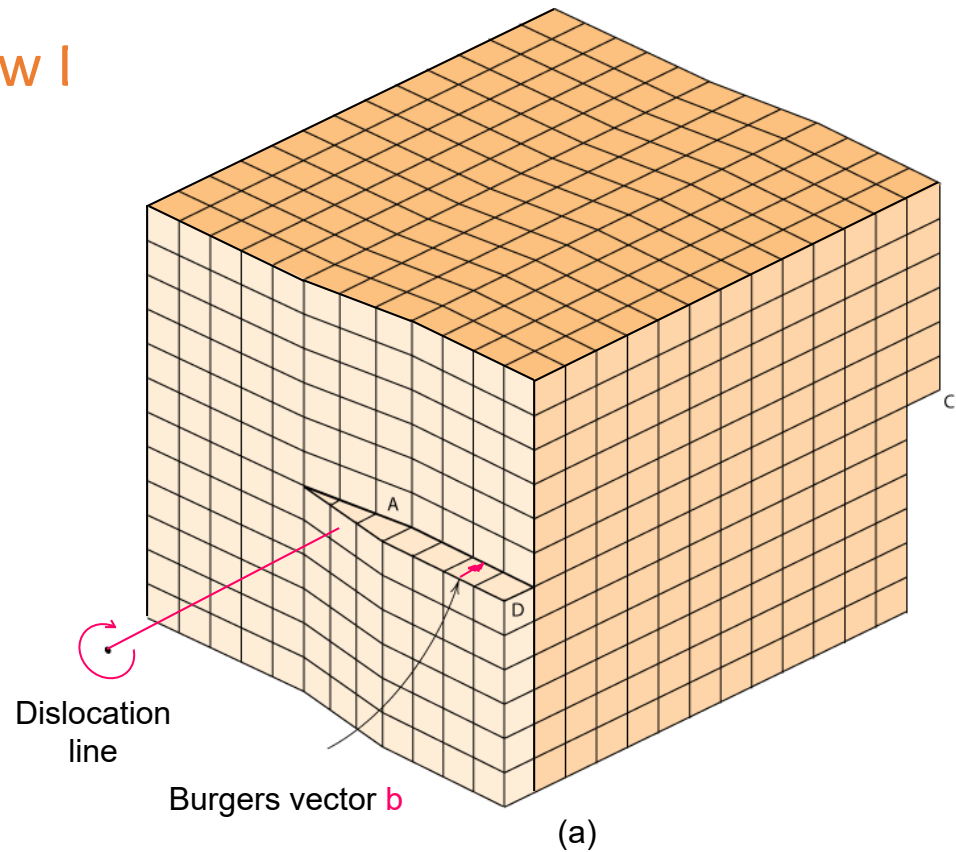


- A useful way to describe a dislocation is to use *Burgers Circuit*:
  - A Burgers Circuit is any atom to atom path taken in a crystal containing dislocations which forms a closed loop.
  - The vector used to close the circuit around a dislocation is called **Burgers Vector,  $\mathbf{b}$** .
    - Burgers vector describes the size and the direction of the main lattice distortion caused by a dislocation

# Imperfections in Solids

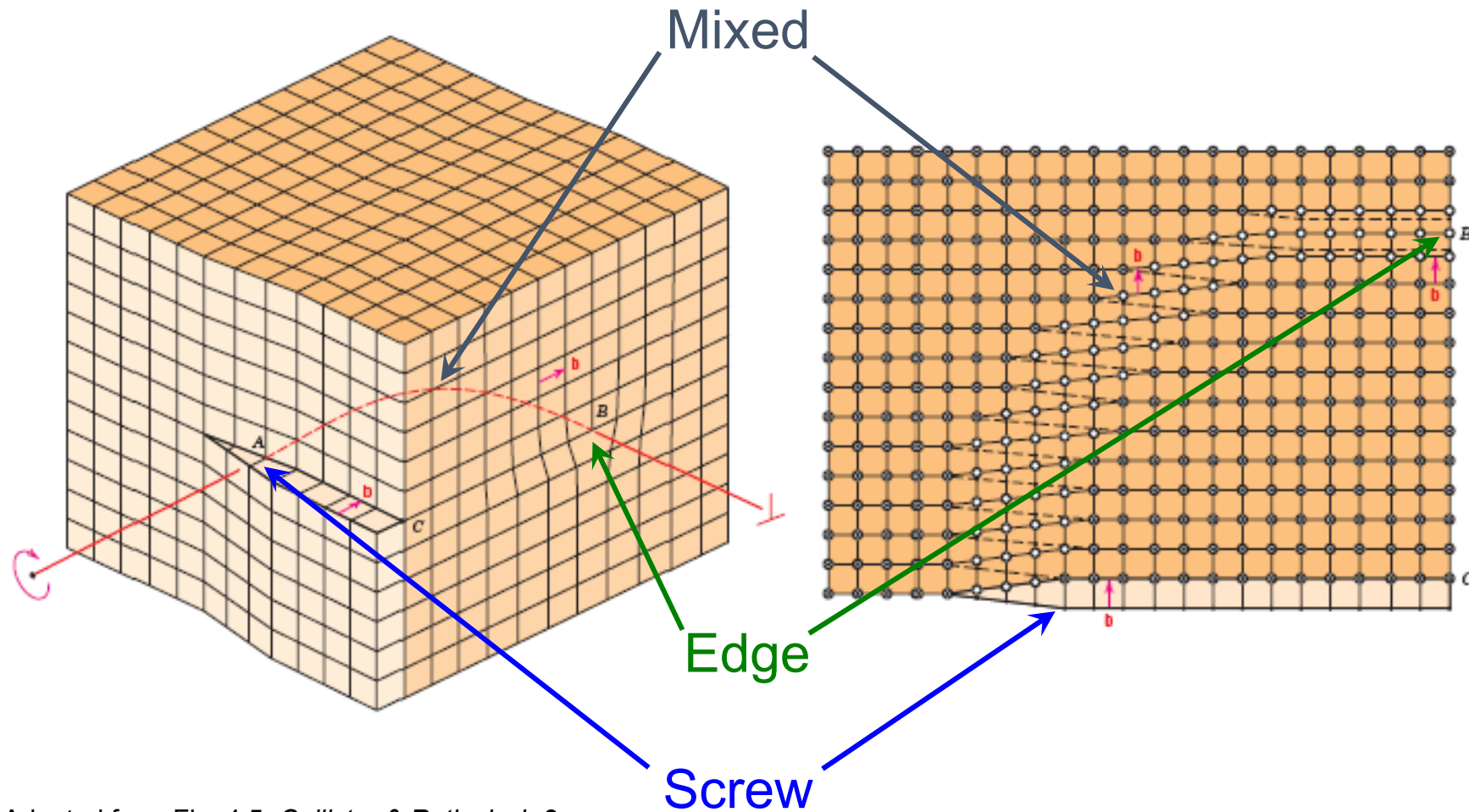
## Screw Dislocation

Screw I



Adapted from Fig. 4.4, *Callister & Rethwisch 8e*.

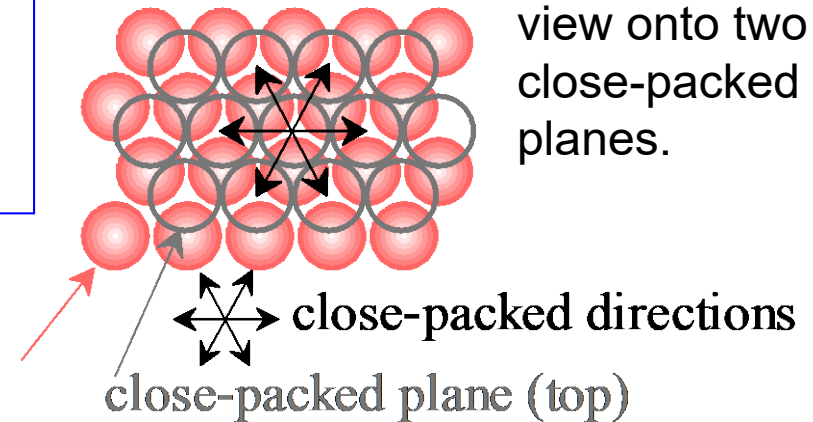
# Edge, Screw, and Mixed Dislocations



Adapted from Fig. 4.5, Callister & Rethwisch 8e.

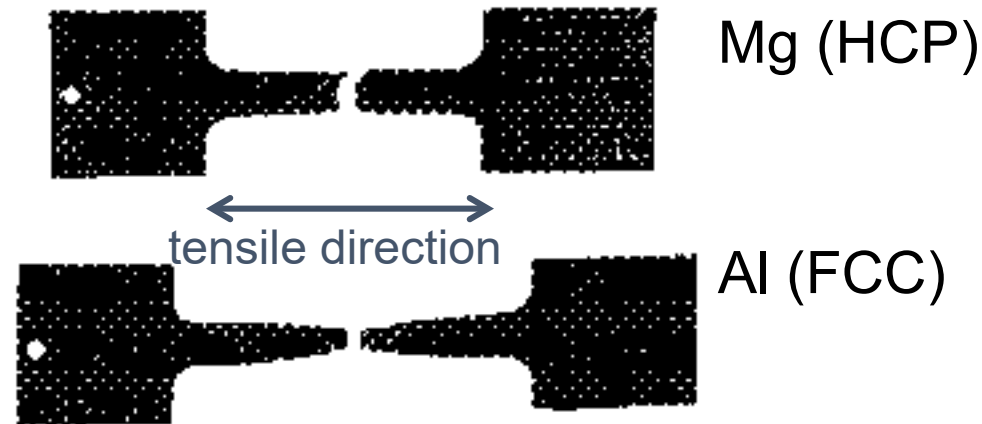
# DISLOCATIONS & CRYSTAL STRUCTURE

- Structure: **close-packed** planes & directions are preferred.



- Comparison among crystal structures:
  - FCC: many close-packed planes/directions; total # is 12
  - HCP: only one plane, 3 directions;
  - BCC: none (special case, # 48)

- Results of tensile testing.



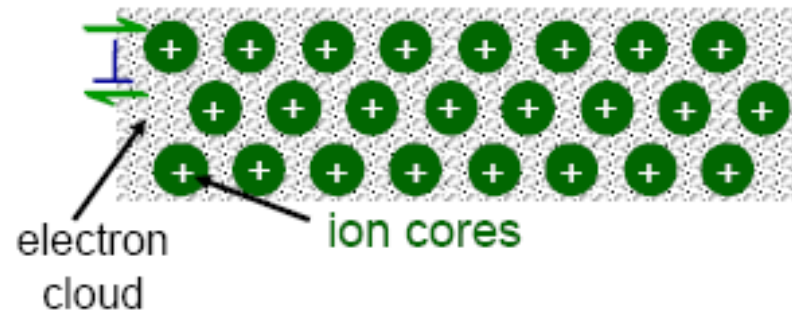
# Dislocations and Materials Strength

- FCC metals are in general more ductile; plastically deform well before failure
- HCP metals are in general less ductile
- BCC metals are stronger due to intersecting slip planes; limited dislocation activity; work harden very quickly

# Dislocations and Materials Strength

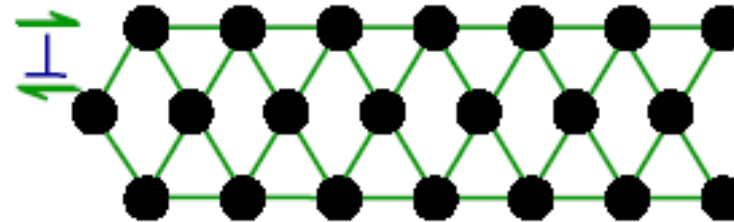
- **Metals:**

Easily form dislocations and allow mobility;  
Not limited with coordination numbers



- **Covalent Ceramics** (Si, diamond):

Remember Covalent Bond !  
How many bonds to break ?  
Finding an equivalent site ?



- **Ionic Ceramics** (NaCl):

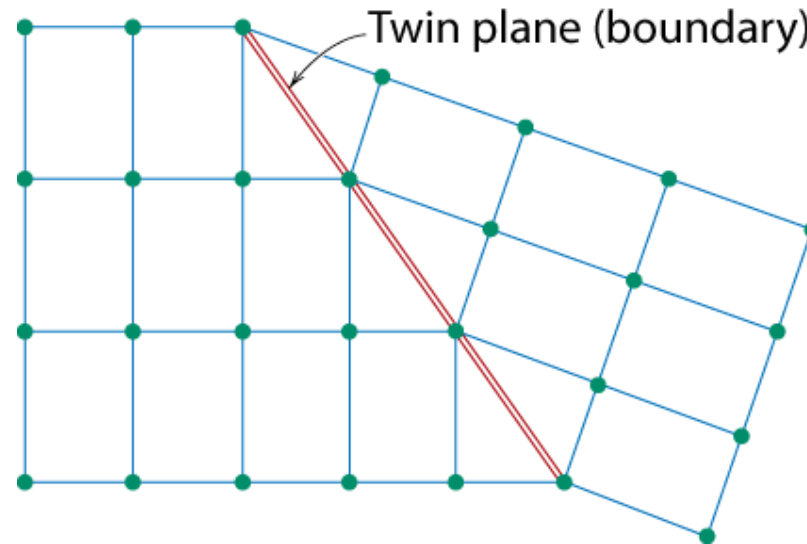
Very large Burgers vector size;  
Finding an equivalent site and overcoming repulsive forces !





# Planar Defects in Solids

- One case is a **twin boundary (plane)**
  - Essentially a reflection of atom positions across the **twin plane**.

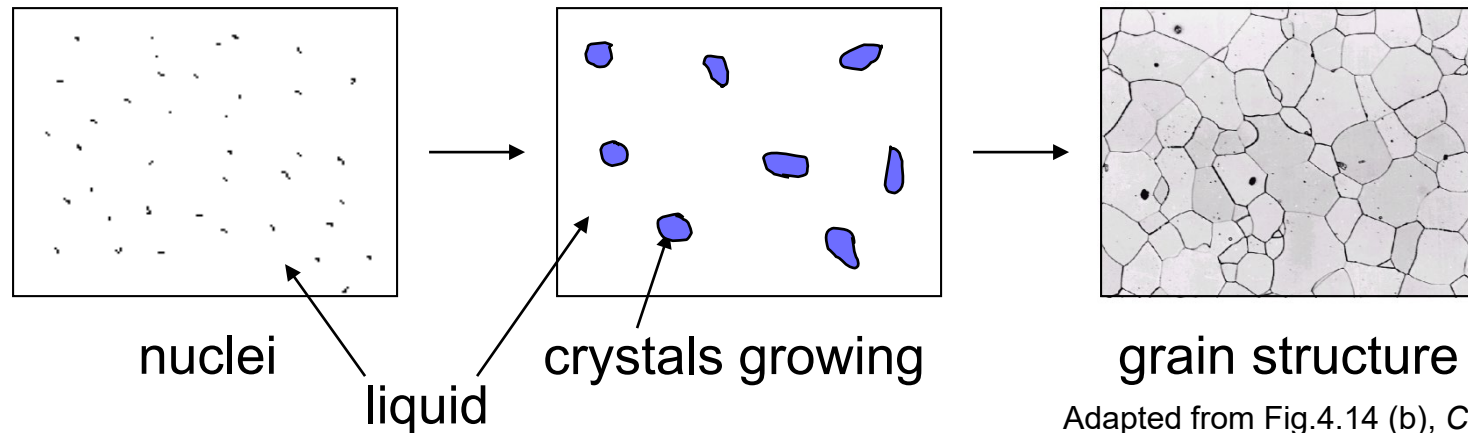


Adapted from Fig. 4.9, *Callister 7e*.

- **Stacking faults**
  - For FCC metals an error in ABCABC packing sequence
  - Ex: ABCABABC

# Imperfections in Solids

- **Solidification**- result of casting of molten material
  - 2 steps
    - Nuclei form
    - Nuclei grow to form crystals – grain structure
  - Start with a molten material – all liquid



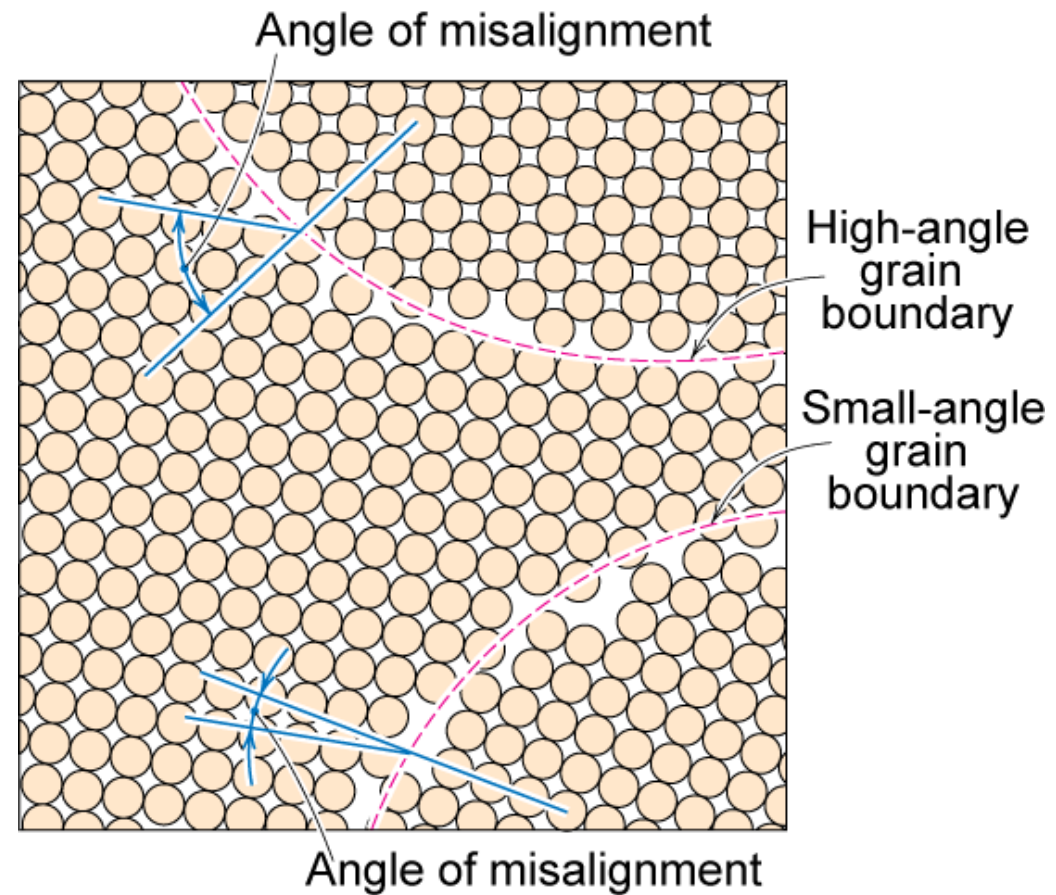
Adapted from Fig.4.14 (b), *Callister 7e*.

- Crystals grow until they meet each other

# Polycrystalline Materials

## Grain Boundaries

- regions between crystals
- transition from lattice of one region to that of the other
- slightly disordered
- low density in grain boundaries
  - high mobility
  - high diffusivity
  - high chemical reactivity



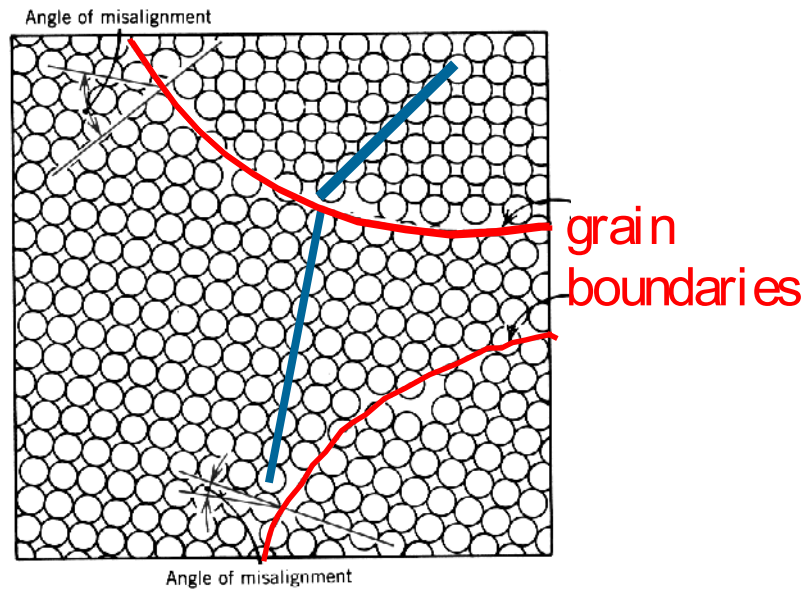
Adapted from Fig. 4.7, *Callister 7e*.

# AREA DEFECTS: GRAIN BOUNDARIES

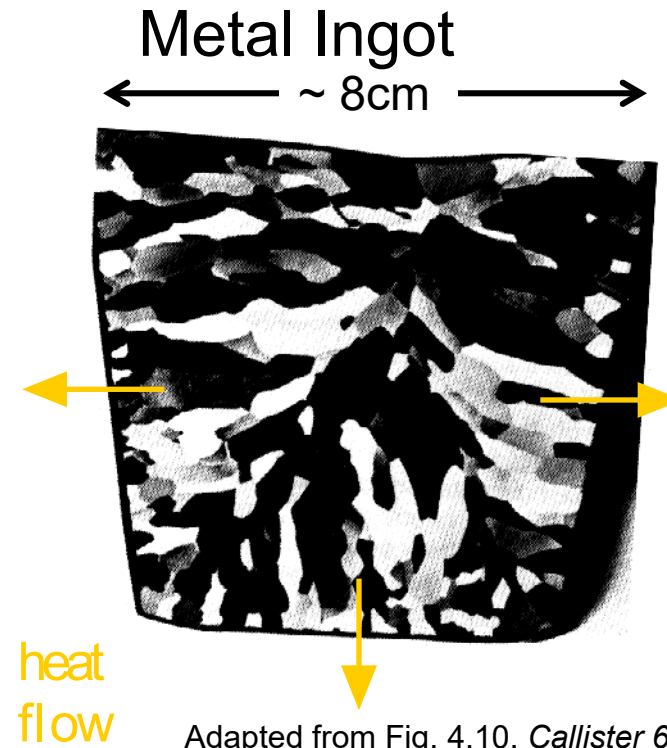
## Grain boundaries:

- are boundaries between crystals.
- are produced by the solidification process, for example.
- have a change in crystal orientation across them.
- impede dislocation motion.

## Schematic



Adapted from Fig. 4.7, *Callister 6e*.



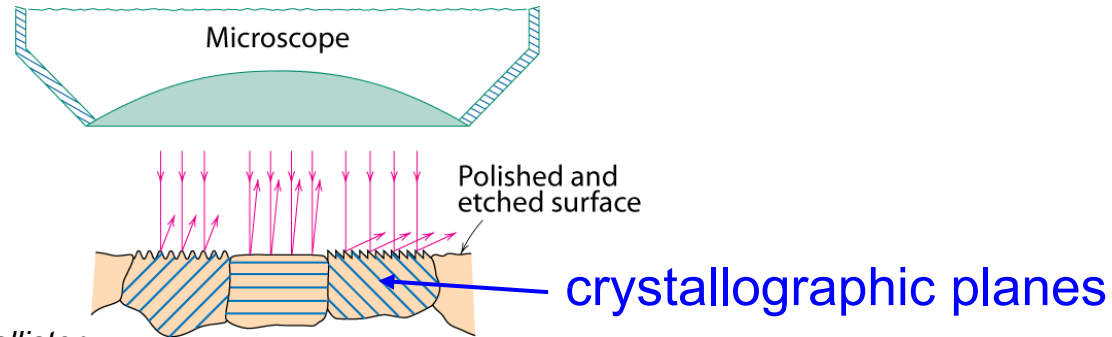
Adapted from Fig. 4.10, *Callister 6e*. (Fig. 4.10 is from *Metals Handbook*, Vol. 9, 9th edition, *Metallography and Microstructures*, Am. Society for Metals, Metals Park, OH, 1985.)

# Microscopic Examination

- Crystallites (grains) and grain boundaries. Vary considerably in size. Can be quite large
  - ex: Large single crystal of quartz or diamond or Si
  - ex: Aluminum light post or garbage can - see the individual grains
- Crystallites (grains) can be quite small (mm or less)
  - necessary to observe with a microscope.

# Optical Microscopy

- Useful up to 2000X magnification.
- Polishing removes surface features (e.g., scratches)
- Etching changes reflectance, depending on crystal orientation.



Adapted from Fig. 4.13(b) and (c), *Callister 7e*. (Fig. 4.13(c) is courtesy of J.E. Burke, General Electric Co.)



Micrograph of  
brass (a Cu-Zn alloy)

← 0.75mm →

# Optical Microscopy

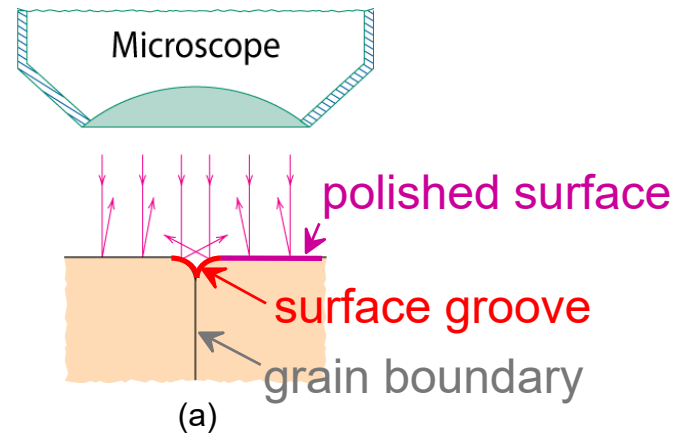
## Grain boundaries...

- are imperfections,
- are more susceptible to etching,
- may be revealed as dark lines,
- change in crystal orientation across boundary.

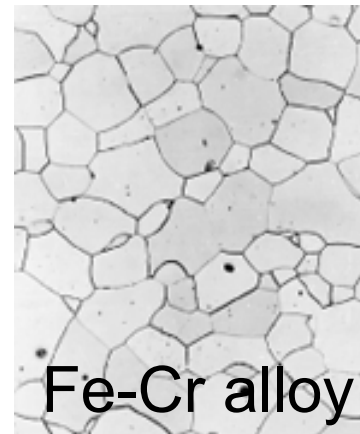
ASTM grain  
size number

$$N = 2^{n-1}$$

number of grains/in<sup>2</sup>  
at 100x  
magnification



(a)



(b)

Adapted from Fig. 4.14(a) and (b), *Callister 7e*.  
(Fig. 4.14(b) is courtesy of L.C. Smith and C. Brady, the National Bureau of Standards, Washington, DC [now the National Institute of Standards and Technology, Gaithersburg, MD].)

# Optical Microscopy

- Polarized light
  - metallographic scopes often use polarized light to increase contrast
  - Also used for transparent samples such as polymers



# Microscopy

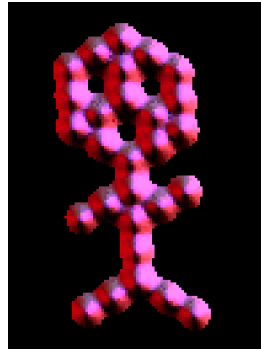
Optical resolution ca.  $10^{-7}$  m = 0.1  $\mu$ m = 100 nm

For higher resolution need higher frequency

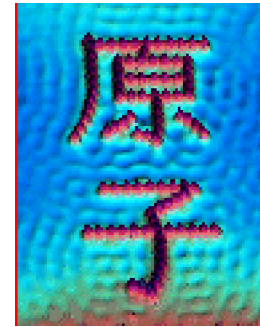
- X-Rays? Difficult to focus.
- Electrons
  - wavelengths ca. 3 pm (0.003 nm)
    - (Magnification - 1,000,000X)
  - Atomic resolution possible
  - Electron beam focused by magnetic lenses.

# Scanning Tunneling Microscopy (STM)

- Atoms can be arranged and imaged!



Carbon monoxide molecules arranged on a platinum (111) surface.



Iron atoms arranged on a copper (111) surface. These Kanji characters represent the word “atom”.

Photos produced from the work of C.P. Lutz, Zeppenfeld, and D.M. Eigler. Reprinted with permission from International Business Machines Corporation, copyright 1995.

# Imperfections in Solids

Dislocations are visible in electron micrographs

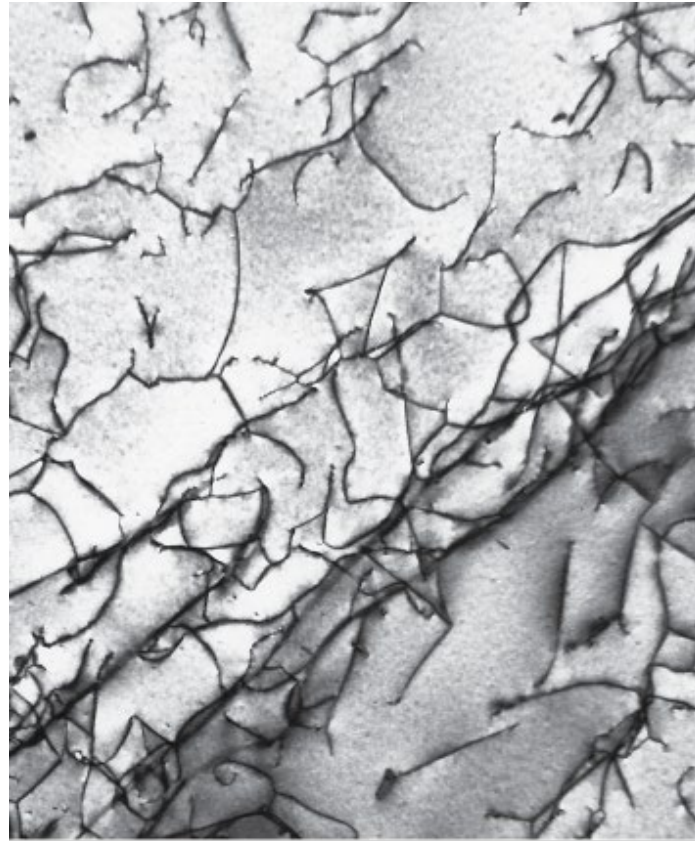


Fig. 4.6, *Callister & Rethwisch 8e.*

0.2  $\mu\text{m}$

# Catalysts and Surface Defects

- A **catalyst** increases the rate of a chemical reaction without being consumed
- Active sites on catalysts are normally surface defects

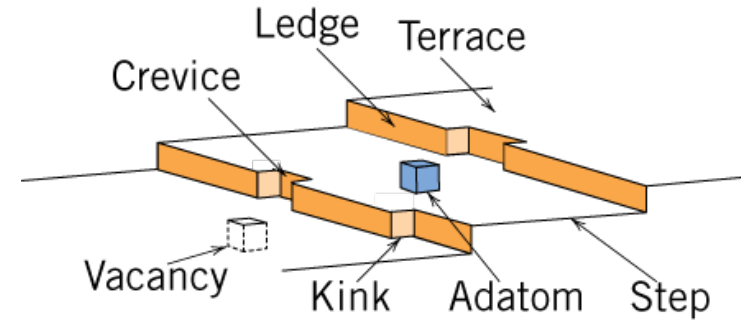


Fig. 4.10, Callister & Rethwisch 8e.

Single crystals of  $(\text{Ce}_{0.5}\text{Zr}_{0.5})\text{O}_2$  used in an automotive catalytic converter

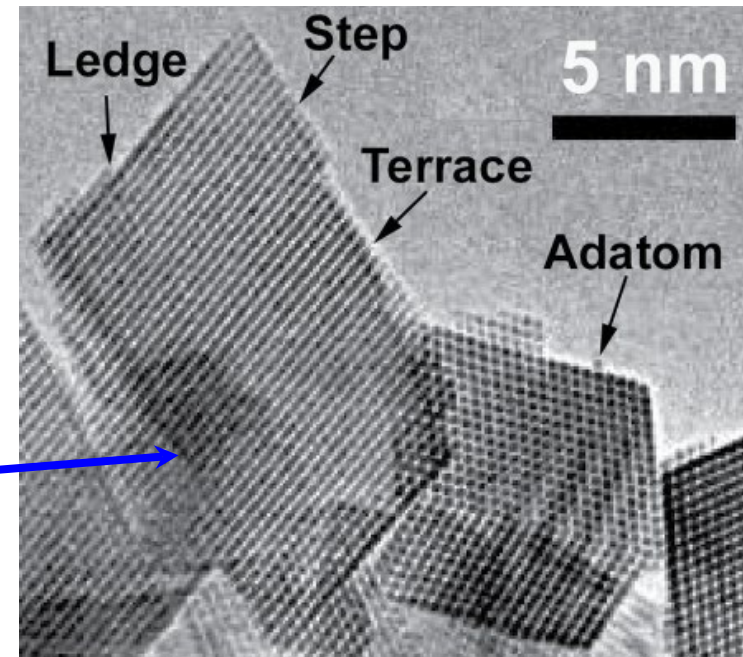
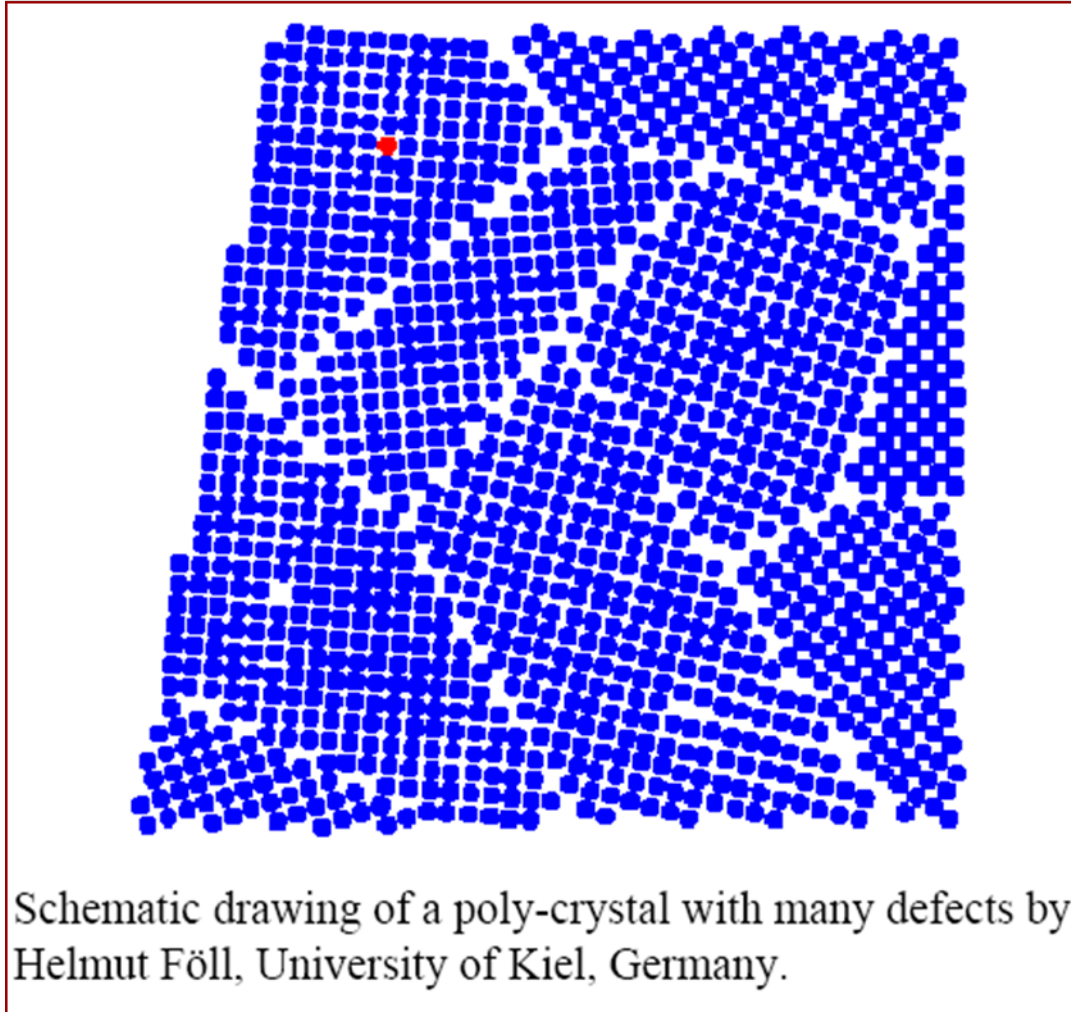


Fig. 4.11, Callister & Rethwisch 8e.

# Microscopic Examination

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- Crystallites (grains) can be quite small (mm or less)
  - necessary to observe with a microscope.

# Imperfections in Solids

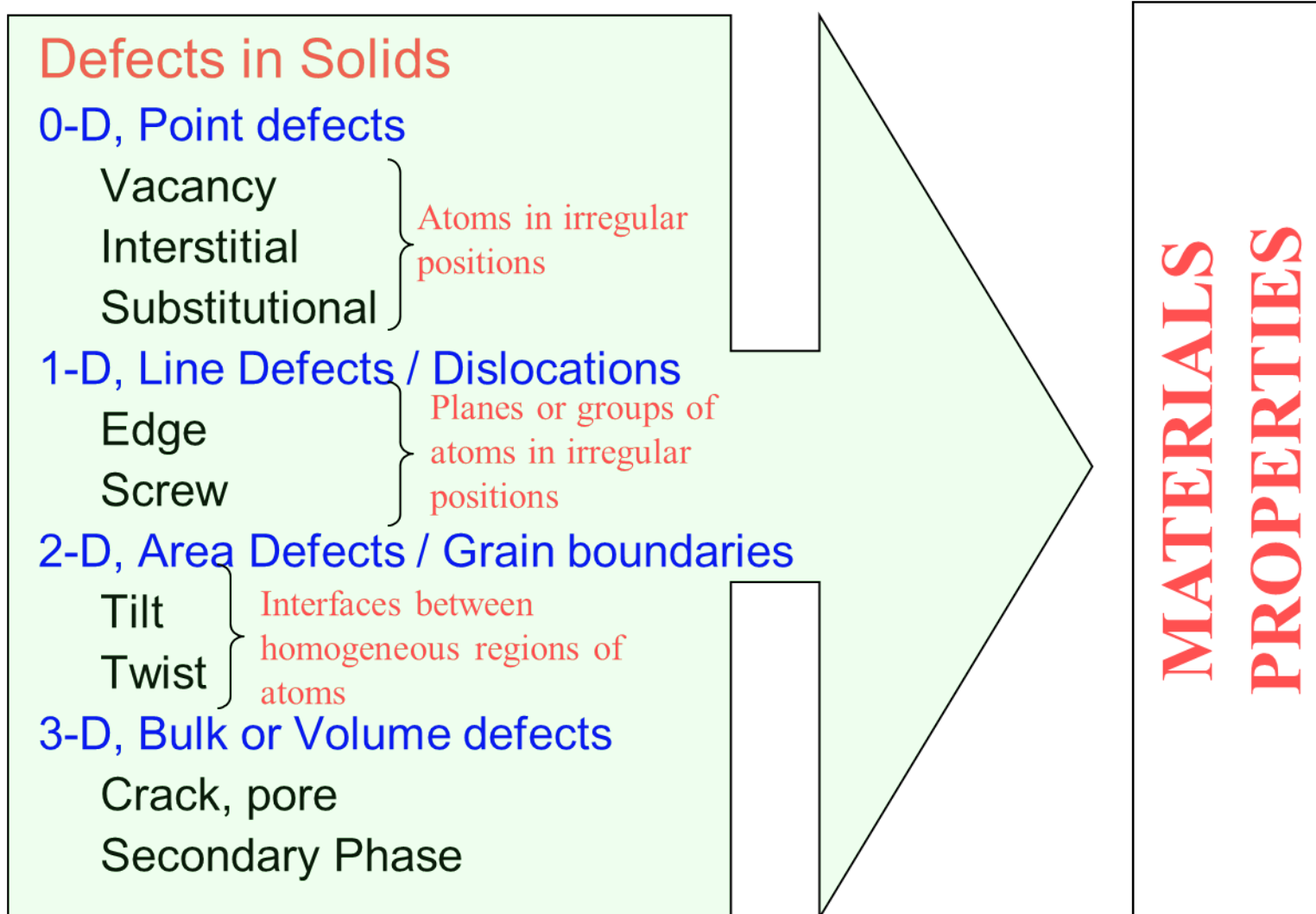


Defects around us:

- A- Color/Price of Precious Stones
- B- Mechanical Properties of Metals
- C- Properties of Semiconductors
- D- Corrosion of Metals

# Imperfections in Solids

Crystals in nature are never perfect, they have *defects* !



# Summary

- **Point**, **Line**, and **Area** defects exist in solids.
- The number and type of defects can be varied and controlled (e.g.,  $T$  controls vacancy conc.)
- Defects affect material properties (e.g., grain boundaries control crystal slip).
- Defects may be desirable or undesirable (e.g., dislocations may be good or bad, depending on whether plastic deformation is desirable or not.)



# Why are we interested IMPERFECTIONS IN SOLIDS ?

“Crystals are like people, it is the defects in them which tend to make them *interesting!*” - Colin Humphreys.

# Measuring Activation Energy

Vacancy density doubles with increasing the temperature from 300 K to 400 K.

What is the activation energy assuming it is not changing with temperature?

$$\rightarrow \frac{N_V}{N} = \exp\left(-\frac{Q_V}{KT}\right), \quad \underline{N_V^{400K} = 2 N_V^{300K}}$$

$$2 = \frac{\frac{N_V^{400K}}{N}}{\frac{N_V^{300K}}{N}} = \frac{\exp\left(-\frac{Q_V}{400K}\right)}{\exp\left(-\frac{Q_V}{300K}\right)}$$

$$\ln(2) = \frac{Q_V}{K} \left( \frac{1}{300} - \frac{1}{400} \right)$$

$$0.693 K = Q_V \left( \frac{1}{300} - \frac{1}{400} \right)$$

← Boltzmann's constant